
Parking Operational Intelligence Platform

Dashboard Development Handoff Guide

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1. Project Objective

The dashboard is NOT a visualization layer.

It is an operational decision support system for:

- Bengaluru Traffic Police
- Smart City Authorities
- Traffic Control Rooms
- Enforcement Teams
- Command Centers

The dashboard consumes exported ML artifacts only.

The dashboard must NEVER:

- retrain models
- rerun clustering
- regenerate embeddings
- recompute forecasts

Everything must be loaded directly from exported files.

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2. Dashboard Architecture

Raw Dataset
↓
Notebook Pipeline
↓
Export Layer
↓

CSV / NPY / GeoJSON
↓
Dashboard

Dashboard is read-only.

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3. Core Deliverables

The dashboard should answer:

Question 1

Where should police deploy next?

Question 2

Which hotspots are getting worse?

Question 3

Which hotspots need towing?

Question 4

What dynamic parking policy
should be applied?

Question 5

Which hotspots behave similarly?

Question 6

What will happen next week?

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4. Required Files

hotspot_master.csv

Primary table.

Contains:

Cell
Archetype
Priority
POI
CIS
Forecast
Trend
Recommendation

Use for:

- rankings
- KPI cards
- hotspot explorer

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hotspots.geojson

Map layer.

Contains:

Geometry
Priority
Forecast
Archetype

Use for:

Leaflet
Mapbox
Folium
DeckGL

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hotspot_timeseries.csv

Daily hotspot history.

Use for:

Trend Charts
Forecast Charts
Historical Analysis

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hotspot_hourly.csv

Hourly patterns.

Use for:

Peak Hour Analysis
Patrol Scheduling

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archetype_profiles.csv

Cluster summaries.

Use for:

Radar Charts
Archetype Cards

---

cluster_map.csv

Cell → Archetype mapping.

Use for:

Map coloring
Cluster filtering

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semantic_hotspot_index.csv

Semantic search database.

Contains:

hotspot_text
priority
archetype
trend

Use for:

Natural Language Search

---

hotspot_embeddings.npy

MiniLM embeddings.

Use for:

Vector Search
Nearest Neighbors
Semantic Retrieval

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hotspot_embedding_lookup.csv

Maps:

Cell

→ Embedding Row

Required for retrieval.

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similarity_graph.csv

Graph relationships.

Use for:

Network Visualizations

---

graph_nodes.csv

Node metadata.

Use for:

Node Coloring
Node Labels
Node KPIs

---

graph_edges.csv

Edge relationships.

Use for:

Graph Rendering
Future GNN

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dashboard_metadata.json

Global KPIs.

Contains:

Rows Processed
Hotspots
Forecast Correlation
Archetypes

Use in header cards.

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5. Dashboard Pages

PAGE 1

Executive Command Center

Audience:

Commissioners
Senior Officers

KPIs

Total Violations

Total Hotspots

Critical Hotspots

Forecasted Risk Zones

Average POI

Average CIS

Charts

Priority Distribution

Bar Chart

Priority Band
vs
Number of Cells

Archetype Distribution

Donut Chart

Night Spillovers

Commercial Arteries

Junction Choke Points

etc.

Trend Distribution

Stacked Bar

Chronic
Growing
Emerging
Stable
Declining

PAGE 2

GIS Operations Map

Most important page.

Source

hotspots.geojson

Layers

Priority Layer

Color:

Green → Yellow → Red

based on:

final_priority_score

CIS Layer

Color:

Blue → Purple

based on:

cis

Archetype Layer

Unique color per archetype.

Forecast Layer

Shows:

Predicted future risk

Clicking hotspot opens:

Cell ID

Archetype

Priority

POI

CIS

Forecast

Trend

Recommended Action

PAGE 3

Hotspot Explorer

Table View

Source:

hotspot_master.csv

Filters

Archetype

Station

Trend

Priority

CIS

Forecast

Columns

Location

Priority

POI

CIS

Forecast

Action

Search

Cell ID

Location

Junction

PAGE 4

Forecast Center

Source:

valid_predictions.csv
hotspot_timeseries.csv

Charts

Actual vs Forecast

Scatter

Forecast Error

Histogram

High Risk Forecasts

Table

Forecast POI
descending

PAGE 5

Archetype Intelligence

Source

archetype_profiles.csv

For each archetype:

Show

Description

Risk

Count

Average CIS

Average POI

Policy Recommendation

Visuals

Radar Chart

Features

POI

CIS

Trend

Spillover

Repeat Rate

Map

Only that archetype.

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Semantic Search

Most innovative page.

Input

Natural Language Query

Examples

Show chronic scooter violations

Busy junction hotspots

Tow ready locations

High spillover commercial areas

Backend

MiniLM Embeddings
+
Cosine Similarity

Output

Top Similar Hotspots

with:

Similarity %

Priority

Archetype

Forecast

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Network Intelligence

Source

graph_nodes.csv
graph_edges.csv

Visualization

Force-directed graph.

Node Size

Priority Score

Node Color

Archetype

Edge Thickness

Similarity

Questions Answered

Which hotspots influence others?

Where does congestion spread?

What communities exist?

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6. Dynamic Parking Policy Engine

This is a dedicated dashboard section.

Not just analytics.

Zone A

CIS < 25

Policy

120 min parking

Base fee

Low enforcement

Zone B

25-50

Policy

60 min parking

1.5x fee

Warning enforcement

Zone C

50-75

Policy

30 min parking

2x fee

Frequent patrols

Zone D

75+

Policy

Tow-ready

Dynamic premium fee

Immediate intervention

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7. Recommended Tech Stack

Frontend

React
TypeScript

Maps

Mapbox GL

or

Leaflet

Charts

Apache ECharts

or

Plotly

Semantic Search

FAISS
+
hotspot_embeddings.npy

Backend

FastAPI

Storage

Artifacts Folder

No database needed initially.

Final Delivery Checklist

The dashboard is complete only when it provides:

✅ Executive Overview

✅ GIS Hotspot Map

✅ Forecast Center

✅ Archetype Intelligence

✅ Semantic Search

✅ Similarity Graph

✅ Dynamic Parking Zone Policies

✅ Downloadable CSV Reports

✅ Exportable PDF Summaries

✅ Zero ML recomputation

If all of the above are implemented, the dashboard becomes a true **Dynamic Parking Zone Intelligence Platform** rather than just a hotspot visualization tool.